The sealed statements of QCALean

These 60 statements are the ones that `lake comparator` checks. Each is followed by the name of its formal version in `verify/Challenge.lean`.

Regions, nets and the lattice

A net over a preordered set R of regions in a C*-algebra A assigns to each region O a star subalgebra $\mathcal{A}(O)$ of A. A C*-algebra is a unital complex C*-algebra, and a star subalgebra is a unital complex star subalgebra. On a lattice, given by an abelian group G of sites, the regions are the subsets of G, ordered by inclusion. For sets O, N of sites,

$$O + N = \{x + y : x \in O, y \in N\}.$$

1. A net over R in a C*-algebra A assigns to each region O a star subalgebra $\mathcal{A}(O)$ of A; an assignment $O \mapsto F(O)$ is a net exactly when it is monotone ($O \leq O'$ implies $F(O) \subseteq F(O')$, isotony) and every $F(O)$ is norm-closed.
`Sealed.net_iff`

2. The lattice causal index and its translations: for an abelian group G, two sets of sites are orthogonal exactly when they are disjoint, and one depends on another exactly when it is contained in it. Translation by $a \in G$ maps a region O to $O+a = \{x : x-a \in O\}$; translations act as automorphisms of the causal index, the product of two automorphisms being their composite and the unit acting as the identity.
`Sealed.lattice_index`

3. The axioms of a net $\mathcal{A}$ over a causal index on R. Generation: the norm closure of the star subalgebra generated by all local algebras is A. Causality: if O_1 and O_2 are orthogonal, every element of $\mathcal{A}(O_1)$ commutes with every element of $\mathcal{A}(O_2)$. Time slice: if O depends on O', then $\mathcal{A}(O) \subseteq \mathcal{A}(O')$. Covariance under a group Γ with homomorphisms κ into the automorphisms of the causal index and α into the star automorphisms of A: $\alpha_\gamma(\mathcal{A}(O)) = \mathcal{A}(\kappa_\gamma(O))$ for every γ and O.
`Sealed.net_axioms_iff`

The CCR algebra and its states

Let V be a real vector space and σ a real bilinear form on V. The Hilbert space $\ell^2(V)$ consists of the square-summable complex functions on the set V. A bilinear form is alternating when $\sigma(u, u) = 0$ for every u. A complex number z satisfies $0 \leq z$ when it is real and non-negative.

4. The Weyl operators and the CCR algebra: for $u \in V$ the Weyl operator $W(u)$ is the bounded operator on $\ell^2(V)$ with

$$(W(u)f)(v) = e^{i\sigma(u,v)/2} f(v + u);$$

the CCR algebra $\mathrm{CCR}(V,\sigma)$ is the norm closure of the star subalgebra generated by the Weyl operators, a C*-algebra; and $W(u)$, regarded as an element of $\mathrm{CCR}(V,\sigma)$, is the same operator.
`Sealed.weyl_and_CCR`

5. For alternating σ every Weyl operator is unitary, $W(u)^* = W(-u)$, and the Weyl relation

$$W(u)W(w) = e^{-i\sigma(u,w)/2}W(u+w)$$

holds.
`Sealed.weyl_relations`

6. Relabelling. Let σ be alternating and $T : V \to V'$ a real linear bijection with $\sigma'(Tu,Tw) = \sigma(u,w)$. The relabelling isomorphism $\mathrm{CCR}(V,\sigma) \to \mathrm{CCR}(V',\sigma')$ maps $W(u)$ to $W(Tu)$ for every u, and two star isomorphisms of $\mathrm{CCR}(V,\sigma)$ onto a C*-algebra that agree on every Weyl operator are equal.
`Sealed.relabel_weyl`

7. States and their characteristic functions: a state of $\mathrm{CCR}(V,\sigma)$ is a continuous complex linear functional ω; a functional is a state exactly when $\omega(1) = 1$ and $\omega(a^*a) \geq 0$ for every a. The characteristic function of a state is $\chi_\omega(u) = \omega(W(u))$.
`Sealed.state_iff`

8. Gaussian states and the state condition: a Gaussian state is a state ω with a symmetric real bilinear form γ (the covariance form) and a real linear functional d (the displacement); a state with such γ and d is Gaussian exactly when

$$\chi_\omega(u) = \exp(i\,d(u) - \gamma(u,u)/4)$$

for every u. A real bilinear form γ satisfies the state condition for σ exactly when $\gamma(u,u) \geq 0$ and $\sigma(u,w)^2 \leq \gamma(u,u)\,\gamma(w,w)$ for all u, w.
`Sealed.gaussianState_iff`

9. Covariance matrices: a phase-space index of n modes is an element of the disjoint union of two copies of $\{1,\dots,n\}$, the positions $q_1,\dots,q_n$ followed by the momenta. The standard symplectic matrix is

$$\Omega = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

in these blocks, and a real $2n \times 2n$ matrix γ is an admissible covariance matrix exactly when it is symmetric and $\gamma - i\Omega$ is positive semidefinite (Hermitian with $\bar{x}^T(\gamma - i\Omega)x \geq 0$ for every complex vector x).
`Sealed.admissible_iff`

The lattice phase space and symbols

Let R be a commutative ring, G an abelian group (the sites) and ι a finite set (the coordinates of one cell). The group algebra $R[G]$ consists of the finitely supported functions $G \to R$ with the convolution product; the element of $R[G]$ with value r at a and 0 elsewhere is written rz^a, and $R[G]$ is the ring of Laurent polynomials in s variables when $G = \mathbb{Z}^s$. The lattice phase space is $R[G]^\iota$: a vector u has a value $u(x) \in R^\iota$ at every site x, nonzero at finitely many sites. A matrix M over $R[G]$, the symbol, acts on it by matrix multiplication.

10. The lattice phase space: its vectors are the ι-tuples of elements of $R[G]$; the value of u at x is the tuple of the coefficients of u_i at x; the support of u is the finite set of sites where this value is nonzero; and translation by a is multiplication by z^a, which shifts values: $(\tau_a u)(x) = u(x-a)$.
`Sealed.lattice_space`

11. Translation-invariant maps and their symbols. An R-linear map L of the lattice phase space is translation invariant when it commutes with every translation; this holds exactly when L is $R[G]$-linear. The $R[G]$-linear maps correspond to the $\iota \times \iota$ matrices over $R[G]$, their symbols, by matrix multiplication, and the symbol of a composite is the product of the symbols (thesis (4.13)). In coordinates (thesis (4.8)), with ℓ_m the coefficient matrix of z^m in the symbol M,

$$(Mu)(x) = \sum_m \ell_m \, u(x-m).$$

`Sealed.symbol`

12. Finite range: the range of a symbol M is the finite set of sites m at which some entry of M has a nonzero coefficient, and

$$\operatorname{supp}(Mu) \subseteq \operatorname{supp} u + \operatorname{range} M$$

for every vector u.
`Sealed.symbol_range`

13. Lifted forms. The involution of $R[G]$ maps rz^a to rz^{-a} (for $G = \mathbb{Z}$, $z \mapsto z^{-1}$); for a matrix M over $R[G]$, $M^{\star T}$ is the transpose of M with the involution applied to every entry. A matrix B over R is regarded as a constant symbol, and it defines the lifted form

$$\tilde{B}(u,w) = \sum_{i,j} B_{ij} \sum_x u_i(x)\, w_j(x)$$

on the lattice phase space. B is alternating when $v^T B v = 0$ for every $v \in R^\iota$.
`Sealed.lifted_form`

14. The twisted criterion (thesis (4.9)–(4.11)): the map of the lattice phase space with symbol M preserves the lifted form of B, $\tilde{B}(Mu, Mw) = \tilde{B}(u,w)$ for all u, w, exactly when $M^{\star T} B M = B$. If moreover $\det B$ is a unit of R, then M is invertible with two-sided inverse $B^{-1} M^{\star T} B$. For every B, the invertible symbols with $M^{\star T} B M = B$ are closed under products and inverses.
`Sealed.twisted_criterion`

Reversible and irreversible cellular automata

Let G be an abelian group, $\mathcal{A}$ a net over the subsets of G in a C*-algebra A, and α a homomorphism from G to the star automorphisms of A, the translations of the algebra. A unital completely positive map of A is a linear map T with $T(1) = 1$ such that every amplification $T \otimes \mathrm{id}_{M_k}$ is positive. The definition of a reversible QCA does not require the inverse of the rule to be local.

15. Reversible quantum cellular automata (thesis Def. 4.1, reversible case): a reversible QCA on $\mathcal{A}$, covariant under α, consists of a star automorphism T of A (the rule) and a finite set N of sites (the neighbourhood); a pair is a reversible QCA exactly when the net is covariant under the translations (acting on regions by $O \mapsto O + a$ and on A by α), T commutes with every α_a, and $T(\mathcal{A}(O)) \subseteq \mathcal{A}(O + N)$ for every region O. The composite of two reversible QCA, first T_1 then T_2, is a reversible QCA with rule $T_2 T_1$ and neighbourhood $N_1 + N_2$ (thesis Cor. 4.3).
`Sealed.reversibleQCA_iff`

16. Quantum cellular automata (thesis Def. 4.1) and the two locality notions of thesis §4.3: for a star-ordered C*-algebra A, a QCA on $\mathcal{A}$, covariant under α, consists of a completely positive map T of A and a finite set N of sites; a pair is a QCA exactly when $T(1) = 1$, the net is covariant under the translations, $\alpha_a T = T\alpha_a$ for every a, and $T(\mathcal{A}(O)) \subseteq \mathcal{A}(O + N)$ for every region O. A map T of A has finite propagation with neighbourhood N (notion I) when $T(\mathcal{A}(O)) \subseteq \mathcal{A}(O+N)$ for every region O, and it factorizes with respect to a set M of sites (notion II) when $T(ab) = T(a)T(b)$ for $a \in \mathcal{A}(O_1)$, $b \in \mathcal{A}(O_2)$ whenever $O_1 + M$ and O_2 are disjoint. Every reversible QCA is a QCA with the same rule and neighbourhood.
`Sealed.qca_iff`

One mode, nearest neighbour

A chain of single modes has the sites $\mathbb{Z}$ and one position and one momentum per site, positions first;

$$\Omega = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

is the one-mode symplectic matrix, also regarded as a constant symbol. A symbol over $\mathbb{Z}$ is a matrix of Laurent polynomials in z, z^m being the element $1z^m$ of $\mathbb{R}[\mathbb{Z}]$; it is symplectic when $M^{\star T}\Omega M = \Omega$, which by the twisted criterion says that its map preserves

$$\sigma(u, w) = \sum_x u(x)^T \Omega w(x).$$

17. One-mode nearest-neighbour rules (thesis Prop. 4.4, corrected). For real 2×2 matrices ℓ_-, ℓ_0, ℓ_+ the nearest-neighbour symbol

$$\hat{L} = \ell_- z^{-1} + \ell_0 + \ell_+ z$$

has the entries $(\ell_-)_{ij} z^{-1} + (\ell_0)_{ij} + (\ell_+)_{ij} z$. It is symplectic, $\hat{L}^{\star T}\Omega\hat{L} = \Omega$, exactly when one of two cases holds: either $\ell_0 = 0$ and the rule is a pure shift, $\ell_+ = 0$ with $\ell_-^T\Omega\ell_- = \Omega$ or $\ell_- = 0$ with $\ell_+^T\Omega\ell_+ = \Omega$ (the case the thesis omits); or $\ell_0^T\Omega\ell_0 = \Omega$, $\ell_+ = \ell_-$, $\det \ell_+ = 0$ (rank at most one) and $\ell_+^T\Omega\ell_0 + \ell_0^T\Omega\ell_+ = 0$.
`Sealed.rev_nn_symplectic_iff`

18. Fourier modes (thesis (4.14)–(4.15)). For a symbol M over $\mathbb{Z}$, the Fourier symbol $\hat{M}(k)$ at $k \in \mathbb{R}$ is M evaluated entrywise on the unit circle at $z = e^{-ik}$:

$$\hat{M}(k)_{ij} = \sum_m c_m e^{-ikm},$$

where c_m is the coefficient of z^m in M_{ij} (a finite sum). If an n-mode symbol M is symplectic, $M^{\star T}\Omega M = \Omega$ with Ω the standard symplectic matrix of n modes, then every Fourier mode is symplectic: $\hat{M}(k)^\dagger \Omega \hat{M}(k) = \Omega$ for every real k (thesis p. 82).
`Sealed.rev_fourierSymbol`

19. The leapfrog symbol of the lattice Klein–Gordon field. Let $m^2 \in \mathbb{R}$. The leapfrog symbol is

$$\hat{F} = \begin{pmatrix} 0 & 1 \\ -1 & z + z^{-1} - m^2 \end{pmatrix}$$

and the inverse leapfrog symbol is

$$\begin{pmatrix} z + z^{-1} - m^2 & -1 \\ 1 & 0 \end{pmatrix}.$$

Both are symplectic and their product (leapfrog times inverse) is 1. The map F of the leapfrog symbol acts on the values $u(x) = (q(x), p(x))$ of a vector of the one-mode chain by

$$(Fu)(x) = (p(x),\ -q(x) + p(x+1) + p(x-1) - m^2 p(x)),$$

the leapfrog step of the lattice Klein–Gordon field, and the trace of its Fourier symbol is $2\cos k - m^2$.

`Sealed.rev_leapfrog_symbol`

The lattice CCR net and reversible Gaussian automata

Let G be an abelian group (the sites) and n the number of modes per site. The lattice phase space V is the lattice phase space over $\mathbb{R}$ whose cell coordinates are the positions and momenta of n modes, positions first; Ω is the standard symplectic matrix of n modes and

$$\sigma(u, w) = \sum_x u(x)^T \Omega w(x)$$

its lifted form. $e_{x,i}$ is the vector with value the i-th unit vector at the site x and 0 elsewhere, and τ_a is translation by a. For a symbol M (a matrix over $\mathbb{R}[G]$) we write $Lu = Mu$ for its map.

20. The lattice CCR net. The phase space V is the lattice phase space over $\mathbb{R}$ with the positions and momenta of n modes as cell coordinates, and σ is the lifted form of Ω, which is alternating; the lattice CCR net is built from V with the form σ, so that it is a net in the CCR algebra $\mathrm{CCR}(V, \sigma)$. The lattice CCR net assigns to a set O of sites the norm closure of the star subalgebra of $\mathrm{CCR}(V, \sigma)$ generated by the Weyl operators $W(u)$ with $\operatorname{supp} u \subseteq O$, and a Weyl operator $W(u)$ lies in $\mathcal{A}(O)$ exactly when $\operatorname{supp} u \subseteq O$. The translation α_a of the CCR algebra maps $W(u)$ to $W(\tau_a u)$. The net is generating, causal, has the time-slice property ($\mathcal{A}(O) \subseteq \mathcal{A}(O')$ whenever O depends on O'; for the lattice index this is isotony, which every net has), and is covariant under the translations.

`Sealed.rev_lattice_ccrNet`

21. Gaussian automata (thesis Def. 4.1, reversible case, and (4.7)). Let M be an n-mode symbol over $\mathbb{R}[G]$ with $M^{\star T} \Omega M = \Omega$ (a symplectic rule). Then there is a reversible QCA on the lattice CCR net, covariant under the translations, whose rule maps $W(u)$ to $W(Lu)$ for every $u \in V$ and whose neighbourhood is the range of M.

`Sealed.rev_rule_qca`

22. The local rule (thesis Lemma 4.2 in Gaussian form). Let M, M' be n-mode symbols over $\mathbb{R}[G]$ with maps L, L', and $e_i = e_{0,i}$ the unit vectors at the site 0. If $Le_i = L'e_i$ for every i, then $M = M'$: a rule is determined by its values on one site. And M is symplectic, $M^{\star T}\Omega M = \Omega$, exactly when these values satisfy the one-cell commutation relations with their translates: $\sigma(Le_i, \tau_x Le_j) = \Omega_{ij}$ if $x = 0$ and 0 otherwise, for all i, j and all $x \in G$.

`Sealed.rev_local_rule`

23. Shifts and on-site rules (thesis p. 70). For every $a \in G$ the translation α_a is the rule of a reversible QCA with neighbourhood $\{a\}$ (the shift). On the chain $G = \mathbb{Z}$, every real $2n \times 2n$ matrix S with $S\Omega S^T = \Omega$ gives a reversible QCA with neighbourhood $\{0\}$ whose rule maps $W(u)$ to $W(L_S u)$, where L_S is the map of the constant symbol S, $(L_S u)(x) = Su(x)$. For one mode, the rotations

$$\begin{pmatrix} \cos\varphi & -\sin\varphi \\ \sin\varphi & \cos\varphi \end{pmatrix}$$

and the squeezings $\mathrm{diag}(e^r, e^{-r})$ satisfy $S\Omega S^T = \Omega$.

`Sealed.rev_examples_onsite`

24. Nearest-neighbour examples on the one-mode chain. For real φ and λ, the example system (4.18) of the thesis, the rotation by φ with symmetric coupling λ, has the nearest-neighbour symbol with ℓ_0 the rotation

$$\begin{pmatrix} \cos\varphi & -\sin\varphi \\ \sin\varphi & \cos\varphi \end{pmatrix}$$

and

$$\ell_+ = \ell_- = \lambda \begin{pmatrix} \sin\varphi & \cos\varphi \\ 0 & 0 \end{pmatrix};$$

it gives a reversible QCA with neighbourhood $\{-1, 0, 1\}$ whose rule maps $W(u)$ to $W(Lu)$. For $m^2 \in \mathbb{R}$ the leapfrog symbol and the inverse leapfrog symbol likewise give reversible QCA with neighbourhood $\{-1, 0, 1\}$, and their rules are inverse to each other.
`Sealed.rev_examples_nn`

Automorphisms of Weyl type and the leapfrog

Cauchy data of the lattice Klein–Gordon field are pairs $d = (d_0, d_1)$ of finitely supported functions $\mathbb{Z} \to \mathbb{R}$; the lattice $\mathbb{Z}^2$ has points $p = (t, x)$, time first, and a test function is a finitely supported function $\mathbb{Z}^2 \to \mathbb{R}$.

25. Automorphisms of Weyl type (the content of thesis (4.7), with the character the thesis omits). Let G be an abelian group of sites, V the lattice phase space with n modes per site and σ the form lifted from Ω. Let T be a star automorphism of $\mathrm{CCR}(V, \sigma)$ and $\Gamma : V \to V$, $c : V \to \mathbb{C}$ arbitrary maps with $T(W(u)) = c(u)W(\Gamma u)$ for every u. Then Γ is a real-linear bijection, it preserves σ, and c is a character of modulus one: $c(u + w) = c(u)c(w)$ and $|c(u)| = 1$.
`Sealed.rev_weyl_type`

26. The leapfrog on Cauchy data. Let $m^2 \in \mathbb{R}$. The Wronskian of Cauchy data is

$$\sigma_S(d, d') = \sum_x d_0(x) d_1'(x) - \sum_x d_1(x) d_0'(x),$$

and the leapfrog F maps (d_0, d_1) to $(d_1, -d_0 + d_1(\cdot + 1) + d_1(\cdot - 1) - m^2 d_1)$. F is a bijection that preserves the Wronskian. There is a real-linear bijection e from Cauchy data onto the one-mode lattice phase space over $\mathbb{Z}$ with $(e\,d)(x) = (d_0(x), d_1(x))$; it carries the Wronskian to σ, $\sigma_S(d, d') = \sigma(e\,d, e\,d')$, and F to the map of the leapfrog symbol: $e(Fd) = \hat{F}(e\,d)$.
`Sealed.rev_cauchy_leapfrog`

27. The lattice Klein–Gordon field and its Cauchy map. Let $m^2 \in \mathbb{R}$. The discrete Klein–Gordon operator is

$$(Ph)(t, x) = h(t + 1, x) + h(t - 1, x) - h(t, x + 1) - h(t, x - 1) + m^2 h(t, x)$$

for $h : \mathbb{Z}^2 \to \mathbb{R}$. For a test function f, the retarded solution $G^+ f$ is the unique h with $Ph = f$ that vanishes at all times $t < t_0$ for some t_0, the advanced solution $G^- f$ the unique h with $Ph = f$ that vanishes at all times $t > t_0$ for some t_0, and the causal propagator is $Ef = G^- f - G^+ f$, a linear map. The phase space of the lattice field is the quotient of the test functions by the kernel of E, and $[f]$ is the class of f in this quotient. The Cauchy map Λ_t at time t sends $[f]$ to $((Ef)(t, \cdot), (Ef)(t + 1, \cdot))$.
`Sealed.rev_lattice_field`

28. The direction of time (thesis (4.7)). Let $m^2 \in \mathbb{R}$, Λ_t the Cauchy map at time t and F the leapfrog. Then $\Lambda_{t+1} = F \circ \Lambda_t$ for every t, and time translation by one step acts on Cauchy data as F^{-1}: if $g(p) = f(p - (1,0))$ for every point p (g is f moved one step to the future), then $F(\Lambda_t[g]) = \Lambda_t[f]$, that is $\Lambda_t[g] = F^{-1}\Lambda_t[f]$, F being bijective.
`Sealed.rev_time_step`

Gaussian states on the lattice

For a state ω of $\mathrm{CCR}(V, \sigma)$ and a star automorphism α, $\omega \circ \alpha$ is the state $x \mapsto \omega(\alpha(x))$. Cells are maps $x \mapsto \gamma_x$ from $\mathbb{Z}$ to real $2n \times 2n$ matrices; they are summable when every entry is absolutely summable over x, symmetric when $\gamma_{-x} = \gamma_x^T$, and of finite range when $\gamma_x = 0$ for $|x| > R$ for some R. On the chain $G = \mathbb{Z}$ the covariance form of cells is

$$\gamma(u, w) = \sum_{x,y} u(x)^T \gamma_{x-y} w(y)$$

and their Fourier series is

$$\hat{\gamma}(k) = \sum_x \gamma_x e^{-ikx}.$$

29. Translation-invariant states. For states of $\mathrm{CCR}(V, \sigma)$: $(\omega \circ \alpha)(x) = \omega(\alpha(x))$; a state is determined by its functional; and a state ω is translation invariant, $\omega \circ \alpha_a = \omega$ for every $a \in G$, exactly when its characteristic function is, $\chi_\omega(\tau_a u) = \chi_\omega(u)$ for every $a \in G$ and $u \in V$.
`Sealed.rev_translation_invariant`

30. The action of a rule on Gaussian data. Let ω be a Gaussian state of $\mathrm{CCR}(V, \sigma)$ with covariance form γ and displacement d, M an n-mode symbol over $\mathbb{R}[G]$ with map L, and T a star automorphism with $T(W(u)) = W(Lu)$ for every u (for symplectic M the rule of its automaton is such a T). Then $\omega \circ T$ is a Gaussian state with covariance form $\gamma(L\cdot, L\cdot)$ and displacement $d \circ L$.
`Sealed.rev_gaussian_evolve`

31. Covariance matrices of finite regions (where the state condition on covariance forms meets the admissibility condition $\gamma - i\Omega \geq 0$ on covariance matrices). Let O be a finite set of sites. The site vectors $e_{x,i}$, $x \in O$, are enumerated by a bijection from pairs (x, i) onto the phase-space indices of $n|O|$ modes that sends (x, q_r) to the position index $e(x, r)$ and (x, p_r) to the momentum index $e(x, r)$ of the same mode, for a bijection e from $O \times \{1, \ldots, n\}$ onto $\{1, \ldots, n|O|\}$; γ_O is the matrix of a form γ on these vectors,

$$(\gamma_O)_{(x,i),(y,j)} = \gamma(e_{x,i}, e_{y,j}),$$

and the covariance matrix of γ on O is $\Omega^T \gamma_O \Omega$, with Ω the standard symplectic matrix of $n|O|$ modes. If γ is symmetric, it satisfies the state condition on the vectors supported in O ($\gamma(u, u) \geq 0$ and $\sigma(u, w)^2 \leq \gamma(u, u)\gamma(w, w)$ whenever $\operatorname{supp} u, \operatorname{supp} w \subseteq O$) exactly when its covariance matrix on O is admissible. The covariance form of every Gaussian state has an admissible covariance matrix on O, and the vacuum form $\sum_x u(x)^T w(x)$ has covariance matrix 1.
`Sealed.rev_covariance_matrix`

32. Cells, their covariance form and their Fourier series. For cells (γ_x), the covariance form is

$$\gamma(u, w) = \sum_{i,j} \sum_{x,y} u_i(x) (\gamma_{x-y})_{ij} w_j(y)$$

(finite sums over the supports), so that

$$\gamma(e_{x,i}, e_{y,j}) = (\gamma_{x-y})_{ij};$$

it is translation invariant, $\gamma(\tau_a u, \tau_a w) = \gamma(u, w)$, and symmetric when the cells are. For summable cells and every real k the matrix series $\sum_x e^{-ikx}\gamma_x$ converges and $\hat{\gamma}(k)$ is its sum; $\hat{\gamma}$ is continuous, and for summable symmetric cells $\hat{\gamma}(k)$ is Hermitian.
`Sealed.rev_cells`

33. The Fourier criterion (thesis Lemma 4.5(i)). Let (γ_x) be summable symmetric cells with covariance form γ and Fourier series $\hat{\gamma}$. The state condition for γ and σ is equivalent to $\gamma + i\sigma \geq 0$: for every finite family $u_1, \ldots, u_m$ of vectors the complex matrix $(\gamma(u_j, u_l) + i\sigma(u_j, u_l))_{jl}$ is positive semidefinite. If $\hat{\gamma}(k) + i\Omega$ is positive semidefinite for every real k, then γ satisfies the state condition; conversely, for cells of finite range, the state condition implies that $\hat{\gamma}(k) + i\Omega$ is positive semidefinite for every k.
`Sealed.rev_fourier_criterion`

34. Saturation for one mode (the matrix content of thesis Lemma 4.5(ii); purity of states is not part of the statement). Let (γ_x) be summable symmetric one-mode cells with $\hat{\gamma}(k) + i\Omega$ positive semidefinite for every real k. Then three conditions are equivalent: $\det \hat{\gamma}(k) = 1$ for every k; $(\Omega\hat{\gamma}(k))^2 = -1$ for every k; and $\hat{\gamma}(k) + i\Omega$ is singular, $\det(\hat{\gamma}(k) + i\Omega) = 0$, for every k.
`Sealed.rev_saturation`

35. The evolution of cells (thesis (4.17), general form). Let M be an n-mode symbol over $\mathbb{Z}$ with map L, ℓ_m the real coefficient matrix of z^m in M, and (γ_x) summable cells with covariance form γ. The cells

$$\gamma'_x = \sum_{m,m'} \ell_m^T \gamma_{x+m-m'} \ell_{m'},$$

the sums running over the range of M (thesis (4.17a)), are summable, symmetric when (γ_x) is, and their covariance form is $\gamma(L\cdot, L\cdot)$; their Fourier series is

$$\hat{\gamma}'(k) = \hat{L}(k)^\dagger \hat{\gamma}(k) \hat{L}(k)$$

for every real k, with $\hat{L}$ the Fourier symbol of M.
`Sealed.rev_evolution_fourier`

Cells and Fourier series on the chain

On the chain $\mathbb{Z}$ with n modes per site, the symbols are matrices over the group algebra $\mathbb{R}[\mathbb{Z}]$ of Laurent polynomials, a vector u of the lattice phase space has a value $u(x) \in \mathbb{R}^{2n}$ at each site, with coordinates $u_i(x)$ indexed by the phase-space index of n modes (positions first), and Ω is the standard symplectic matrix. For a Laurent polynomial p write p_m for its coefficient of z^m.

36. Cells, their forms, Fourier series and pull-backs. Let $\gamma = (\gamma_x)_{x\in\mathbb{Z}}$ be a family of real $2n \times 2n$ matrices. It is *cell data* exactly when every entry is absolutely summable,

$$\sum_x |(\gamma_x)_{ij}| < \infty,$$

and $\gamma_{-x} = \gamma_x^T$ for every x. Its form is

$$\gamma(u, w) = \sum_{i,j} \sum_{x,y} u_i(x)\, (\gamma_{x-y})_{ij}\, w_j(y)$$

(a finite sum). Its Fourier series $\hat{\gamma}(k)$ is the sum of the series $\sum_x e^{-ikx}\gamma_x$ of complex matrices when the series is summable, which it is for cell data, and 0 by convention otherwise. The on-site cells of a matrix g_0 are g_0 at the origin and 0 elsewhere. For a matrix C of Laurent polynomials, of any size, the coefficient matrix c_m of z^m has the entries $(C_{ki})_m$, the range of C is the finite set of m with $c_m \neq 0$, and the pull-back cells of a family ν along C are

$$(C_*\nu)_d = \sum_{m,m'} c_m^T \, \nu_{d+m-m'} \, c_{m'},$$

the sum over m, m' in the range of C. For a square matrix M of Laurent polynomials (a symbol) the coefficient matrix ℓ_m of z^m likewise has the entries the coefficients of z^m, $(\ell_m)_{ij} = (M_{ij})_m$, and the evolved cells are $M_*\gamma$. The Fourier symbol of a square matrix M of Laurent polynomials at the wave number k is M evaluated entrywise at $z = e^{-ik}$, $p \mapsto \sum_m p_m e^{-ikm}$. Finally, the symplectic form σ of the lattice phase space V, the form from which the lattice CCR net and the CCR algebra $\mathrm{CCR}(V, \sigma)$ are built, is the form lifted from Ω,

$$\sigma(u, w) = \sum_{i,j} \Omega_{ij} \sum_x u_i(x) \, w_j(x).$$

`Sealed.dyn_cells`

The example system

From here on the chain carries one mode per site, so that cells and Fourier symbols are real or complex 2×2 matrices in the coordinates (q, p), written in blocks of 1×1 matrices, and

$$\Omega = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

For real φ write

$$R_\varphi = \begin{pmatrix} \cos\varphi & -\sin\varphi \\ \sin\varphi & \cos\varphi \end{pmatrix}$$

for the rotation by φ and

$$\Gamma_\pm = \begin{pmatrix} \sin\varphi & \cos\varphi \\ 0 & 0 \end{pmatrix}.$$

37. The example system (thesis (4.18)). For real φ and λ, the example rule $L_{\varphi,\lambda}$ is the symbol with the coefficient matrices $\ell_0 = R_\varphi$, $\ell_1 = \ell_{-1} = \lambda\Gamma_\pm$ and $\ell_m = 0$ for $|m| \geq 2$. For every wave number k, its Fourier symbol is the real matrix $\hat{L}(k) = R_\varphi + 2\lambda \cos k \, \Gamma_\pm$, and $\hat{L}(-k) = \hat{L}(k)$, $\det \hat{L}(k) = 1$,

$$\operatorname{tr} \hat{L}(k) = 2(\cos\varphi + \lambda \cos k \sin\varphi)$$

and $\hat{L}(k)^T \Omega \hat{L}(k) = \Omega$. A symbol M on the chain with n modes per site is symplectic when $M^{\star T} \Omega M = \Omega$, and every power $L_{\varphi,\lambda}^t$, $t \in \mathbb{N}$, is symplectic.

`Sealed.dyn_example_system`

38. The regime (thesis (4.19)–(4.20)). For real φ and λ, the pair (φ, λ) lies in the regime exactly when $|\cos\varphi| + |\lambda \sin\varphi| < 1$ and $\lambda \sin\varphi \neq 0$. The first condition holds exactly when every mode is elliptic, $|\operatorname{tr} \hat{L}(k)| < 2$ for every real k, and it implies $\sin\varphi \neq 0$. For $\sin\varphi \neq 0$ it is equivalent to $|\lambda| < f_{\mathrm{crit}}$ with $f_{\mathrm{crit}} = (1 - |\cos\varphi|)/|\sin\varphi|$ (4.20), and $f_{\mathrm{crit}} = \tan(\varphi/2)$ for $0 < \varphi \leq \pi/2$,

$$f_{\mathrm{crit}} = \cot(\varphi/2) = 1/\tan(\varphi/2)$$

for $\pi/2 < \varphi < \pi$.

`Sealed.dyn_regime`

39. The rotation angle, the powers of the symbol and the spectral projectors (thesis (4.19), Lemma 4.6 and (4.21)–(4.24)). For real φ, λ and k let $\theta(k) = \arccos(\cos\varphi + \lambda\cos k\sin\varphi)$, with the real arccosine, which takes values in $[0,\pi]$ and is extended by 0 above 1 and by π below -1; let

$$N(k) = (\sin\theta(k))^{-1}(\cos\theta(k)\,1 - \hat{L}(k)),$$

a real matrix, with the convention $0^{-1} = 0$ (in the regime $\sin\theta(k) > 0$), and let

$$P_k = \frac{1}{2}(1 + iN(k)),$$

a complex matrix with complex conjugate $\bar{P}_k$. The angle θ is even and continuous. In the regime: on $[0,\pi]$, θ is strictly increasing if $\lambda\sin\varphi > 0$ and strictly decreasing if $\lambda\sin\varphi < 0$; and for every k, $\cos\theta(k) = \cos\varphi + \lambda\cos k\sin\varphi$, $\operatorname{tr}\hat{L}(k) = 2\cos\theta(k)$, $0 < \theta(k) < \pi$, θ is differentiable at k with $\theta'(k) = \lambda\sin\varphi\sin k/\sin\theta(k)$,

$$\hat{L}(k)^t = \frac{\sin t\theta(k)}{\sin\theta(k)}\,\hat{L}(k) - \frac{\sin(t-1)\theta(k)}{\sin\theta(k)}\,1$$

and

$$\hat{L}(k)^t = \cos t\theta(k)\,1 - \sin t\theta(k)\,N(k)$$

for every $t \in \mathbb{N}$, $N(k)^2 = -1$, $P_k + \bar{P}_k = 1$, $P_k^2 = P_k$, $P_k\bar{P}_k = \bar{P}_kP_k = 0$,

$$\hat{L}(k) = e^{i\theta(k)}P_k + e^{-i\theta(k)}\bar{P}_k,$$

and

$$\hat{L}(k)^t = e^{it\theta(k)}P_k + e^{-it\theta(k)}\bar{P}_k$$

for every $t \in \mathbb{N}$.

`Sealed.dyn_angle_projectors`

40. The invariant form (thesis (4.28)). For real φ, λ and k let

$$\hat{\varepsilon}(k) = -\frac{\operatorname{sgn}(\sin\varphi)}{\sin\theta(k)}\,\Omega\,(\cos\theta(k)\,1 - \hat{L}(k)),$$

with sgn the sign function ($\operatorname{sgn}0 = 0$) and the convention $1/0 = 0$ for the quotient (in the regime $\sin\theta(k) > 0$). For all parameters, $\hat{\varepsilon}(-k) = \hat{\varepsilon}(k)$ and $\hat{L}(k)^T\hat{\varepsilon}(k)\hat{L}(k) = \hat{\varepsilon}(k)$. In the regime, $\hat{\varepsilon}$ is a continuous function of k, and for every k:

$$\hat{\varepsilon}(k) = \frac{\operatorname{sgn}(\sin\varphi)}{\sin\theta(k)}\begin{pmatrix}\sin\varphi & -\lambda\cos k\sin\varphi\\ -\lambda\cos k\sin\varphi & \sin\varphi - 2\lambda\cos k\cos\varphi\end{pmatrix};$$

$\hat{\varepsilon}(k)$ is symmetric and positive definite, $\det\hat{\varepsilon}(k) = 1$ and $(\Omega\hat{\varepsilon}(k))^2 = -1$;

$$\hat{\varepsilon}(k) = \operatorname{sgn}(\sin\varphi)\,i\Omega\,(P_k - \bar{P}_k);$$

and for every real g, the complex matrix $g\hat{\varepsilon}(k) + i\Omega$ is positive semidefinite exactly when $g \geq 1$.

`Sealed.dyn_invariant_form`

41. The stationary covariance cells of the example system (thesis Theorem 4.8, with the completeness direction, which goes beyond the thesis). Let (φ,λ) lie in the regime and let (γ_x) be cell data on the chain with one mode per site that are reflection symmetric, $\gamma_{-x} = \gamma_x$ for every x. Then the following are equivalent: (i) the cells are stationary, $(L_{\varphi,\lambda})_*\gamma = \gamma$, and satisfy the Fourier state condition, $\hat{\gamma}(k) + i\Omega \geq 0$ (positive semidefinite) for every real k; (ii) there is a continuous function $g : \mathbb{R} \to \mathbb{R}$ with $g(-k) = g(k)$ and $g(k) \geq 1$ for every k such that $\hat{\gamma}(k) = g(k)\,\hat{\varepsilon}(k)$ for every k.

`Sealed.dyn_stationary_iff`

42. The lattice Klein–Gordon leapfrog and its inverse have a mode that is not elliptic, for every real m^2; for $m^2 = 0$ the inverse is the example rule at $(\pi/2, 1)$, on the boundary of the regime. For real m^2 let F be the leapfrog symbol of the lattice Klein–Gordon field with mass squared m^2, and F^{-1} the symbol with $FF^{-1} = 1$. At every wave number k the Fourier symbols of F and F^{-1} have the trace $2\cos k - m^2$, and for each of them some mode is not elliptic: there is a k with $|\operatorname{tr} \hat{F}(k)| \geq 2$, and one with $|\operatorname{tr} \widehat{F^{-1}}(k)| \geq 2$. For $m^2 = 0$, F^{-1} is the example rule $L_{\pi/2,1}$, and $(\pi/2, 1)$ is not in the regime, lying on its boundary:

$$|\cos\frac{\pi}{2}| + |1 \cdot \sin\frac{\pi}{2}| = 1.$$

`Sealed.dyn_leapfrog`

Convergence

The Fourier conventions are those of the section on Gaussian states on the lattice,

$$\hat{\gamma}(k) = \sum_x e^{-ikx}\gamma_x.$$

Integrals over $[-\pi, \pi]$ are Lebesgue integrals for the Lebesgue measure, and the limit of a sequence $(a_t)_{t\in\mathbb{N}}$ is taken as $t \to \infty$.

43. The Riemann–Lebesgue lemma along a phase. Let $s \subseteq \mathbb{R}$ be measurable and let $\theta : \mathbb{R} \to \mathbb{R}$ be injective on s and, at every $x \in s$, differentiable within s with derivative $\theta'(x) \neq 0$. Then for every $h : \mathbb{R} \to \mathbb{C}$, $\int_s h(k)\, e^{ir\theta(k)}\, dk \to 0$ as the real number r tends to ∞. The integral is the Lebesgue integral, which takes the value 0 for a non-integrable integrand, so that the statement has content when h is integrable on s.

`Sealed.dyn_riemann_lebesgue`

44. The limit covariance (thesis Prop. 4.7). For real φ, λ and a family (γ_x) of real 2×2 matrices, the limit matrix is

$$C(k) = \frac{1}{2}\big(\hat{\gamma}(k) + N(k)^T\hat{\gamma}(k)N(k)\big),$$

and the limit cells are the real parts

$$\gamma_{\infty,x} = \operatorname{Re}\frac{1}{2\pi}\int_{-\pi}^{\pi} e^{ikx}\, C(k)\, dk,$$

entrywise. Let (φ, λ) lie in the regime. For every k, $C(k)$ is invariant, $\hat{L}(k)^T C(k)\hat{L}(k) = C(k)$, and if $\hat{\gamma}(k) + i\Omega$ is positive semidefinite, so is $C(k) + i\Omega$. If (γ_x) are cell data, then every entry of $k \mapsto e^{ikx}C(k)$ is integrable on $[-\pi, \pi]$, the integral $\frac{1}{2\pi}\int_{-\pi}^{\pi} e^{ikx}\, C(k)\, dk$ is real and hence equal to $\gamma_{\infty,x}$, and $\gamma_{\infty,-x} = \gamma_{\infty,x}^T$ for every x; if moreover the cells are reflection symmetric, $\gamma_{-x} = \gamma_x$, and $\hat{\gamma}(k) + i\Omega \geq 0$ at the wave number k, then $C(k) = g\,\hat{\varepsilon}(k)$ for a real number $g \geq 1$ (the form of thesis Theorem 4.8 at this wave number).

`Sealed.dyn_limit_covariance`

45. Convergence of the covariance (thesis Prop. 4.7). Let (γ_x) be cell data on the chain with one mode per site and $\gamma^{(t)} = (L_{\varphi,\lambda *})^t\gamma$ the cells evolved t times. For all real φ, λ, k and $t \in \mathbb{N}$,

$$\hat{\gamma}^{(t)}(k) = (\hat{L}(k)^t)^T\, \hat{\gamma}(k)\, \hat{L}(k)^t.$$

If (φ, λ) lies in the regime, then for every site x and every entry, $(\gamma_x^{(t)})_{ab}$ converges to the corresponding entry of the limit cell $\gamma_{\infty,x}$, and for all vectors u, w of the lattice phase space the forms converge, $\gamma^{(t)}(u, w) \to \gamma_\infty(u, w)$, where γ_∞ is the form of the limit cells.

`Sealed.dyn_tendsto_cells`

46. Without coupling the covariance need not converge. There are reflection-symmetric cell data (γ_x) on the chain with one mode per site satisfying the Fourier state condition, $\hat{\gamma}(k) + i\Omega \geq 0$ for every k, such that for the uncoupled example rule $L_{\pi/2,0}$ (a quarter turn at every site, $\lambda = 0$) the position-position entry of the evolved cell at the origin, $(\gamma_0^{(t)})_{qq}$ with $\gamma^{(t)} = (L_{\pi/2,0*})^t \gamma$, does not converge as $t \to \infty$.

`Sealed.dyn_no_coupling`

47. Convergence of the states (the weak-* part of thesis Theorem 4.9). Let (φ, λ) lie in the regime, let (γ_x) be cell data on the chain with one mode per site, and let ω be a Gaussian state of the CCR algebra of the lattice phase space with covariance form the form of (γ_x) and displacement 0. For $t \in \mathbb{N}$ let α_t be the star automorphism of the CCR algebra with $\alpha_t(W(u)) = W(L^t_{\varphi,\lambda} u)$ (the automorphism of the symplectic rule $L^t_{\varphi,\lambda}$), and $\omega_t = \omega \circ \alpha_t$ the evolved Gaussian state. Then the characteristic functions converge pointwise, $\omega_t(W(u)) \to \exp(-\gamma_\infty(u,u)/4)$ for every u, with γ_∞ the form of the limit cells, and there is a Gaussian state ω_∞ with covariance form γ_∞ and displacement 0 such that $\omega_t(a) \to \omega_\infty(a)$ for every element a of the CCR algebra. (The symplecticity of the powers, supplied as a hypothesis, always holds.)

`Sealed.dyn_limit_state`

Gaussian channels and irreversible automata on the chain

For real bilinear forms β, β' on a real vector space V, the condition $\beta + i\beta' \geq 0$ means that for every finite family $u_1, \ldots, u_m$ in V the complex matrix $(\beta(u_j, u_k) + i\beta'(u_j, u_k))_{jk}$ is positive semidefinite. A completely positive map of a C*-algebra is a linear map all of whose matrix amplifications are positive. On the chain, τ_a is the translation of the lattice phase space by $a \in \mathbb{Z}$, the lattice CCR net assigns to $O \subseteq \mathbb{Z}$ the local algebra $\mathcal{A}(O)$, and the translations act on the CCR algebra by the star automorphisms α_a with $\alpha_a(W(u)) = W(\tau_a u)$. The range N_M of a symbol M is the set of m at which some entry has a nonzero coefficient. Statements about lattice channels concern the channels with the given data; the existence of a Gaussian channel with prescribed data is not asserted.

48. Gaussian channels (thesis §4.3). Let σ be a real bilinear form on a real vector space V. A Gaussian channel on $\mathrm{CCR}(V, \sigma)$ consists of a linear map $L : V \to V$, a bilinear form ν (the noise), a linear form μ (the displacement) and a completely positive map f of the CCR algebra; such a quadruple is a Gaussian channel exactly when ν is symmetric, the channel condition (4.34) $\nu + i(\sigma - \sigma(L\cdot, L\cdot)) \geq 0$ holds, and

$$f(W(u)) = \exp(i\mu(u) - \nu(u,u)/4)\, W(Lu)$$

for every $u \in V$. For bilinear forms β, β' on V, the condition $\beta + i\beta' \geq 0$ is the condition on Gram matrices stated at the head of this section. The concatenation $T_2 \circ T_1$ of two Gaussian channels has the map $a \mapsto f_2(f_1(a))$, the linear map $L_2 L_1$, the noise $\nu_1 + \nu_2(L_1\cdot, L_1\cdot)$ and the displacement $\mu_1 + \mu_2 \circ L_1$.

`Sealed.dyn_gaussian_channel`

49. Lattice channels (thesis p. 92) are quantum cellular automata. Let T be a Gaussian channel on the CCR algebra of the lattice phase space of the chain with n modes per site, M a symbol and (ν_x) a family of real $2n \times 2n$ matrices. T is a lattice channel with rule M and noise cells (ν_x) exactly when its linear map is $u \mapsto Mu$, its noise is the form of (ν_x), the family (ν_x) is cell data, and its displacement is translation invariant, $\mu(\tau_a u) = \mu(u)$ for all a and u. A lattice channel commutes with the translations, $\alpha_a \circ T = T \circ \alpha_a$, and there is a quantum cellular automaton on the lattice CCR net, covariant under the translations, with rule T and neighbourhood the range N_M.

`Sealed.dyn_lattice_channel`

50. Locality of lattice channels (thesis Lemma 4.11, in both directions). Let M be a symbol on the chain with n modes per site. The pull-back cells of the on-site cells $\delta_{\cdot,0}\Omega$ along M are the coefficient matrices of $M^{\star T}\Omega M$, and their form is $\sigma(M\cdot, M\cdot)$. Let T be a lattice channel with rule M and noise cells (ν_x). (I) For every set $N \subseteq \mathbb{Z}$, T has finite propagation with neighbourhood N exactly when $N_M \subseteq N$. (II) For every finite set $M' \subseteq \mathbb{Z}$ with $0 \in M'$ and $-x \in M'$ for every $x \in M'$, T factorizes with respect to M' exactly when $\nu_x = 0$ for every $x \notin M'$ and the range of $M^{\star T}\Omega M$ is contained in M' (the cells of $\sigma(M\cdot, M\cdot)$ vanish outside M'; thesis (4.36)).
`Sealed.dyn_locality`

51. Factorization does not imply finite propagation, and the converse separation for the rule 0 holds for every lattice channel with the given data (thesis Lemma 4.10(v)); the existence of such a channel is not proved. Let $n \geq 1$ be the number of modes per site. (a) There is a Gaussian channel that is a lattice channel with the shift as rule (the symbol $z \cdot 1$) and noise cells 0, that factorizes with respect to $\{0\}$, and that does not have finite propagation with neighbourhood $\{0\}$. (b) For every finite set $M' \subseteq \mathbb{Z}$ there are cell data (ν_x) of finite range, $\nu_m = 0$ for $|m| > R$ for some $R \in \mathbb{N}$, such that the rule 0 and the noise ν satisfy the channel condition $\nu + i(\sigma - \sigma(0\cdot, 0\cdot)) \geq 0$, and no lattice channel with rule 0 and noise cells (ν_x) factorizes with respect to M'. (That such a channel exists rests on the characterization of Gaussian channels by their data, thesis Thm. 2.6, which is not asserted here.)
`Sealed.dyn_independence`

52. Concatenation of lattice channels (thesis p. 96). If T_1 and T_2 are lattice channels on the chain with n modes per site, with rules L_1, L_2 and noise cells ν_1, ν_2, then $T_2 \circ T_1$ is a lattice channel with rule L_2L_1 and noise cells $\nu_1 + L_{1*}\nu_2$. For all symbols L, L_1, L_2 and families ν, ν_1, ν_2 of real $2n \times 2n$ matrices: the form of $L_*\nu$ is $\nu(L\cdot, L\cdot)$, with ν also denoting the form of the family;

$$N_{L_2L_1} \subseteq N_{L_1} + N_{L_2};$$

if $\nu_x = 0$ outside a finite set A, then $(L_*\nu)_d = 0$ for $d \notin A + (N_L - N_L)$; the range of $L^{\star T}\Omega L$ is contained in $N_L - N_L$; if N_1, N_2 are finite sets, N_2 nonempty, $N_{L_i} \subseteq N_i$ and $\nu_{i,x} = 0$ for $x \notin N_i - N_i$ $(i = 1, 2)$, then $N_{L_2L_1} \subseteq N_1 + N_2$ and $\nu_1 + L_{1*}\nu_2$ vanishes outside $(N_1 + N_2) - (N_1 + N_2)$; and for on-site cells $\delta_{\cdot,0}a$, $\delta_{\cdot,0}b$ of real matrices a, b, the cells $\delta_{\cdot,0}a + L_{1*}(\delta_{\cdot,0}b)$ vanish off the origin exactly when $L_{1*}(\delta_{\cdot,0}b)$ does (4.39). The thesis writes the ranges of the composite as equalities; the statement gives the inclusions.
`Sealed.dyn_concatenation`

53. The example system with noise: two steps with on-site noise (thesis p. 97, corrected) and a lossy chain. Let $r = (\sin\varphi, \cos\varphi)^T$, and for a real 2×2 matrix ν_0 let $\delta_{\cdot,0}\nu_0$ be the on-site cells. (a) The pull-back cell at distance two is

$$(L_{\varphi,\lambda *}\delta_{\cdot,0}\nu_0)_2 = (\nu_0)_{qq}\,\lambda^2\, rr^T.$$

(b) If ν_0 is positive semidefinite, $(L_{\varphi,\lambda *}\delta_{\cdot,0}\nu_0)_x = 0$ for every $x \neq 0$ exactly when $\lambda = 0$ or $(\nu_0)_{qq} = 0$. By (4.39) these pull-back cells decide whether the noise of two steps with on-site noise ν_0 stays on site, and the claim of the thesis that for nonzero coupling it does not holds for isotropic noise and fails for $\mathrm{diag}(0, b)$: (c) for $\lambda \neq 0$ and $c > 0$ the cell at distance two for $\nu_0 = c\,1$ is nonzero; (d) for $\nu_0 = \mathrm{diag}(0, b)$ with $b \geq 0$ the pull-back cells vanish off the origin, for every λ. (e) The lossy chain: for $0 < \eta < 1$, the rule $\sqrt{\eta}\,L_{\varphi,\lambda}$ and the on-site noise cells $\delta_{\cdot,0}(1-\eta)1$ satisfy the channel condition

$$\nu + i(\sigma - \sigma(\sqrt{\eta}L\cdot, \sqrt{\eta}L\cdot)) \geq 0;$$

the range of $(\sqrt{\eta}L)^{\star T}\Omega(\sqrt{\eta}L)$ is contained in $\{0\}$; the rule $\sqrt{\eta}\,L_{\varphi,\lambda}$ is not symplectic; and for $\lambda \neq 0$ the noise cells $\nu + (\sqrt{\eta}L)_*\nu$ of two steps are nonzero at distance 2.
`Sealed.dyn_example_noise`

54. The Fourier form of the channel condition (thesis (4.34)) for noise of finite range. Let M be a symbol on the chain with n modes per site and (ν_x) cell data of finite range, $\nu_m = 0$ for $|m| > R$. Then the rule M and the noise ν (the form of the cells) satisfy the channel condition $\nu + i(\sigma - \sigma(M\cdot, M\cdot)) \geq 0$ exactly when the matrix $\hat{\nu}(k) + i\Omega - i\,\hat{M}(k)^\dagger \Omega \hat{M}(k)$ is positive semidefinite for every real k, where $\hat{M}(k)$ is the Fourier symbol of M and † the conjugate transpose.
`Sealed.dyn_fourier_condition`

55. Dilations on the chain at the level of covariances (thesis Lemma 4.12, with (4.40)). Let S be a matrix of Laurent polynomials for $n + m$ modes per site, the n system modes first, with

$$S^{\star T}(\Omega_n \oplus \Omega_m)S = \Omega_n \oplus \Omega_m;$$

let S_{11} be its system-to-system block and S_{21} its system-to-ancilla block (rows of the ancilla, columns of the system), and let γ_0 be a real $2m \times 2m$ matrix with $\gamma_0 + i\Omega_m \geq 0$. Then the rule S_{11} and the noise of the cells $\nu = S_{21*}(\delta_{\cdot,0}\gamma_0)$ satisfy the channel condition

$$\nu + i(\sigma - \sigma(S_{11}\cdot, S_{11}\cdot)) \geq 0;$$

$$\hat{\nu}(k) = \hat{S}_{21}(k)^\dagger\, \gamma_0\, \hat{S}_{21}(k)$$

for every k, with $\hat{S}_{21}(k)$ the entrywise evaluation at $z = e^{-ik}$; $N_{S_{11}} \subseteq N_S$; and $\nu_d = 0$ for $d \notin N_S - N_S$.
`Sealed.dyn_dilation`

Examples

These statements show that some hypotheses of the theorems above can be met (the regime, cell data with the Fourier condition, noiseless lattice channels) and that the objects are not degenerate. No Gaussian state is exhibited, and no lattice channel with noise.

56. The pure-shift case is not empty and is not covered by the other case: the symbol with $\ell_- = 1$ and $\ell_0 = \ell_+ = 0$ (the shift z^{-1}) is symplectic, while $\ell_0 = 0$ does not satisfy $\ell_0^T \Omega \ell_0 = \Omega$. The other case contains symplectic nearest-neighbour symbols with $\ell_0 \neq 0$ and $\ell_+ = \ell_- \neq 0$.
`Sealed.witness_rev_nn_shift`

57. The vacuum cells meet the Fourier condition: the one-mode cells $\gamma_0 = 1$ and $\gamma_x = 0$ for $x \neq 0$ are summable and symmetric, $\hat{\gamma}(k) = 1$, and $\hat{\gamma}(k) + i\Omega$ is positive semidefinite for every real k.
`Sealed.witness_rev_vacuum_fourier`

58. An automaton that is not on-site: there is a reversible QCA on the one-mode chain whose rule does not map the local algebra $\mathcal{A}(\{0\})$ of the site 0 into itself.
`Sealed.witness_rev_not_onsite`

59. The hypotheses on parameters and cells can be met: the pair $(\pi/2, 1/2)$ lies in the regime, with $f_{\mathrm{crit}} = 1$ for $\varphi = \pi/2$; and the vacuum cells $\delta_{\cdot,0}\,1$ on the chain with one mode per site are reflection-symmetric cell data with constant Fourier series $\hat{\gamma}(k) = 1$, for which $\hat{\gamma}(k) + i\Omega$ is positive semidefinite for every k.
`Sealed.witness_dyn_regime_vacuum`

60. Lattice channels exist for the example rule: for all real φ, λ there is a Gaussian channel T on the CCR algebra of the chain with one mode per site that is a lattice channel with rule $L_{\varphi,\lambda}$ and noise cells 0, with $T(W(u)) = W(L_{\varphi,\lambda}u)$ for every u.
`Sealed.witness_dyn_lattice_channel`